

Corporate Average Fuel Economy Standards

Updated October 5, 2026

Congressional Research Service

<https://crsreports.congress.gov>

R49361



R49361

October 5, 2026

Naseeb A. Souweidane
Analyst in Transportation
Policy

Corporate Average Fuel Economy Standards

The National Highway Traffic Safety Administration (NHTSA) administers Corporate Average Fuel Economy (CAFE) standards that regulate the fuel consumption of vehicles. These standards require automotive manufacturers to produce vehicles with greater fuel-efficiency with the stated goal to reduce energy consumption and limit dependence on foreign oil. Title 49, Section 32902, of the U.S. Code sets several statutory obligations for CAFE standards, including a mandate to set the maximum feasible standards for each model year and considerations to be used in making that determination. The stringency of CAFE standards has generally increased since their implementation in 1978.

NHTSA sets separate CAFE standards each model year for light-, medium-, and heavy-duty vehicles. Light-duty vehicles are separated into three fleet or compliance categories: (1) domestic passenger vehicles, (2) imported passenger vehicles, and (3) light trucks. Manufacturers are required to meet CAFE standards for each fleet in which they produce vehicles. NHTSA expresses CAFE standards through a mathematical function represented as a curve. The CAFE curve sets a range for vehicles in a given fleet each model year, and the specific standards are dependent on footprint, a vehicle attribute that is a measurement of vehicle size.

NHTSA and the Environmental Protection Agency (EPA) both have roles in the enforcement of CAFE standards. EPA verifies automotive manufacturers' CAFE data, conducts tests to measure the fuel economy and emissions of vehicles, calculates manufacturers' CAFE values, determines the average CAFE values, and issues final CAFE reports to manufacturers and NHTSA. NHTSA then identifies whether the fleet performance is greater than or equal to the standards set for a manufacturer's fleet. Additionally, manufacturers are required to label certain fuel economy information on vehicles, and EPA and NHTSA coordinate to administer the labeling requirements.

NHTSA offers several flexibilities for manufacturers to comply with CAFE standards. These flexibilities include credits from overcompliance with CAFE standards, fuel consumption improvement values (FCIVs), and incentives and limitations in calculations for alternative fuels. Credits for overcompliance with standards can be (with some limitations) applied to previous or future model years, transferred to another fleet owned by a manufacturer, or traded among manufacturers. The CAFE program also allows manufacturers to boost their compliance through FCIVs derived from certain air conditioning and off-cycle technologies. Alternative fuel vehicles include electric, ethanol, biodiesel, and natural gas vehicles are incorporated into CAFE standards through calculations with certain incentives and limitations.

NHTSA generally administers the CAFE program, and Congress may reevaluate the program and agency authorities. Members of Congress might choose to introduce bills that would redefine vehicle fleet classifications or alter what the Secretary of Transportation must consider when determining the maximum feasible average fuel economy. Additionally, the stringency of these standards has changed over time, and Congress may further adjust them. Congress may explore aspects of the CAFE program's compliance and enforcement. For example, Congress may consider EPA's involvement in the enforcement of the program.

In addition, the civil penalty associated with the CAFE program is NHTSA's primary mechanism for enforcing CAFE standards, and it is currently set at \$0.00. Due to this reduction to \$0.00, many aspects of these CAFE standards may no longer be as applicable as before since manufacturers are no longer incentivized to comply with these standards. Congress may choose to reset the civil penalty amount. If Congress does not reset the civil penalty, Congress may choose to still engage in activities related to CAFE standards if Members anticipate future changes to the civil penalty or wish to continue to oversee aspects of the CAFE program, such as monitoring fuel efficiency of vehicle fleets and maintaining vehicle fuel economy labeling.

Congress may reevaluate the flexibilities that NHTSA offers manufacturers to comply with CAFE standards. Credits that manufacturers receive for overcompliance with their CAFE program obligations may be altered by legislation, such as through provisions that would set limitations on banking or trading credits. Congress could also reevaluate NHTSA and EPA's considerations of FCIVs. Congress has modified the incentives for alternative fuel vehicles through several laws and could choose to further modify incentives through new legislation. Congress may choose to amend NHTSA's statutory limitations on the consideration of certain alternative fuel vehicles when setting CAFE standards. Congress could also choose to make no changes to the CAFE program and allow relevant agencies to determine various aspects of the program.

Contents

Introduction	1
CAFE Standards Background.....	2
Statutory Obligations for Standard Setting	3
Stringency of CAFE Standards	5
CAFE Targets	5
Compliance.....	6
Credits	6
Civil Penalty.....	7
Labeling.....	9
Fuel Consumption Improvement Values (FCIVs)	10
Alternative Fuel Vehicles	10
Selected Policy Options.....	12
Vehicle Fleet Definitions.....	13
Standard Setting	13
Role of EPA.....	14
Compliance Enforcement.....	15
Flexibilities.....	15
Credits	15
FCIVs.....	16
Alternative Fuels.....	16

Figures

Figure 1. Civil Penalties Collected by NHTSA from 1985 to 2025	9
--	---

Contacts

Author Information.....	18
-------------------------	----

Introduction

Vehicle fuel economy is regulated at the federal level by the National Highway Traffic Safety Administration (NHTSA). *Fuel economy* is defined in the *U.S. Code* as “the average number of miles traveled by an automobile for each gallon of gasoline (or equivalent amount of other fuel) used.”¹ In the United States, fuel economy is usually expressed as miles per gallon (mpg). A vehicle’s average mpg generally signals the number of miles driven per gallon of fuel (e.g., gasoline, diesel, or alternative fuel) consumed based on certain assumptions, such as city or highway driving. Federal policymakers regulate vehicle fuel economy to limit fuel consumption and promote various goals, such as improving energy security and saving consumers money on fuel.² NHTSA’s fuel economy targets for automotive manufacturers are called Corporate Average Fuel Economy (CAFE) standards. Congress has shaped aspects of the CAFE program through legislation, including by addressing the enforcement of the program, stringency of the CAFE standards, and flexibilities provided to meet these standards.³ Lawmakers may further consider the CAFE standards and aspects of the program.

CAFE standards set a target fuel economy that a manufacturer must reach on average for vehicles produced and sold in a given model year. NHTSA sets different standards for vehicle compliance categories, also known as “fleets,” depending on a vehicle’s weight and whether it was manufactured domestically or imported. NHTSA determines whether manufacturers meet their fuel economy targets for their fleets in a model year. If a manufacturer exceeds the target fuel economy for its fleet, it may receive credits. If a manufacturer does not meet its target fuel economy, it must use credits to cover the shortfall or pay a civil penalty if it does not have credits.

Congress and the U.S. Department of Transportation (DOT) have changed CAFE standards since their inception in the 1970s. For example, Congress and DOT have increased the stringency of mpg targets for vehicles, altered how NHTSA enforces these standards, and provided flexibilities to automotive manufacturers to achieve these standards. Automotive manufacturers have increased the net fuel economy of their fleets since the inception of CAFE standards.

Regulatory changes have required automotive manufacturers to adjust the portfolios of vehicles they produce and sell in the United States to comply with CAFE standards. Reportedly, some stakeholders in the automotive industry have expressed concerns about frequent regulatory changes at the federal level, noting that they affect long-term product planning.⁴ In 2024, the Biden Administration issued a final rule updating CAFE standards for model years 2027-2031; the rule generally increased the stringency of mpg targets.⁵ Through the FY2025 reconciliation law (P.L. 119-21), Congress reduced the civil penalties for violations of CAFE standards to \$0. In 2026, the Trump Administration finalized a rule that amended the CAFE standards updated by the

¹ 49 U.S.C. §32901(a)(11).

² Department of Transportation, “Corporate Average Fuel Economy (CAFE) Standards,” updated August 11, 2014, <https://www.transportation.gov/mission/sustainability/corporate-average-fuel-economy-cafe-standards>.

³ P.L. 94-163; P.L. 110-140; P.L. 103-272; and P.L. 119-21.

⁴ Larry P. Vellequette, “Auto Industry Confidence Index: Automakers See Future Prospects Degrading,” *Automotive News*, August 17, 2025, <https://www.autonews.com/manufacturing/automakers/an-auto-industry-confidence-index-automakers-0817/>; and Lindsay VanHulle and John Irwin, “Tariffs, Changing EV Demand Add Volatility to Automotive Product Planning,” *Automotive News*, July 20, 2025, <https://www.autonews.com/manufacturing/an-tariffs-policy-impact-product-planning-0721/>.

⁵ National Highway Traffic Safety Administration (NHTSA), “Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond,” 89 *Federal Register* 52540, June 24, 2024.

Biden Administration. The Trump Administration final rule reduced the stringency and addressed other aspects of the standards.⁶

This report covers CAFE standards, particularly for light-duty vehicles and NHTSA's regulation of CAFE standards.⁷ Other federal and state emissions standards that may relate to the CAFE program are not included in this report.⁸ This report also provides a brief history of CAFE standards and covers several components of the CAFE program affected by recent congressional and regulatory actions. These components include the statutory obligations of the CAFE program, the stringency of CAFE standards, the process of determining CAFE targets, compliance and enforcement of these standards, and flexibilities afforded to alternative fuel vehicles. The report concludes with options that Congress may consider if it chooses to address policy issues concerning CAFE standards.

CAFE Standards Background

Congress enacted the Energy Policy and Conservation Act (EPCA; P.L. 94-163) in 1975, which required DOT to administer vehicle fuel economy standards known as CAFE standards. CAFE standards required manufacturers to produce vehicles with greater fuel-efficiency with the stated goal to reduce energy consumption and limit dependence on foreign oil.⁹ The Secretary of Transportation delegates to NHTSA the authority to regulate the average mpg fuel consumption of vehicles in an automotive manufacturer's fleet using CAFE standards.¹⁰ Vehicles produced by a manufacturer in a given model year are separated into different fleets.¹¹ NHTSA sets CAFE standards for all light-duty vehicles each model year, separated into three fleets, or compliance categories. The light-duty vehicle compliance categories are (1) domestic passenger vehicles, (2) imported passenger vehicles, and (3) light trucks.¹² Manufacturers producing vehicles in multiple compliance categories are to meet the CAFE standard for each category. The CAFE standard set

⁶ NHTSA, "The Safer Affordable Fuel-Efficient (SAFE) Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks," 91 *Federal Register* 61988, September 30, 2026.

⁷ NHTSA "includes all vehicles with a gross vehicle weight rating of 8,500 pounds or less" in the light-duty Corporate Average Fuel Economy (CAFE) program. NHTSA, "CAFE Public Information Center," accessed April 7, 2026, <https://www.nhtsa.gov/corporate-average-fuel-economy/cale-public-information-center>.

⁸ For more on the Environmental Protection Agency (EPA) emissions regulations and state emissions regulations, see CRS Report R48168, *California and the Clean Air Act (CAA) Waiver: Frequently Asked Questions*, by Benjamin M. Barczewski, Kathryn G. Kynett, and Emily N. Peterson.

⁹ P.L. 94-163.

¹⁰ The Secretary of Transportation's CAFE authority is delegated to the NHTSA administrator. See 49 C.F.R. §1.95(a) and 49 C.F.R. §1.95(j) and 49 U.S.C. §32902.

¹¹ 49 U.S.C. §32901(a)(16) defines *model year* as "the annual production period of a manufacturer, as decided by the [NHTSA] Administrator, that includes January 1 of that calendar year; or that calendar year if the manufacturer does not have an annual production period." 40 C.F.R. §85.2304(a) defines the *annual production period* as "either: when any vehicle or engine within the engine family is first produced; or on January 2 of the calendar year preceding the year for which the model year is designated, whichever date is later. The annual production period ends either: When the last such vehicle or engine is produced; or on December 31 of the calendar year for which the model year is named, whichever date is sooner." 40 C.F.R. §85.2302 defines model year as "the manufacturer's annual production period (as determined under §85.2304) which includes January 1 of such calendar year, provided, that if the manufacturer has no annual production period, the term 'model year' shall mean the calendar year."

¹² 49 C.F.R. §523.4 defines *passenger automobile* as "any automobile (other than an automobile capable of off-highway operation) manufactured primarily for use in the transportation of not more than 10 individuals. A medium-duty passenger vehicle that does not meet the criteria for non-passenger motor vehicles in § 523.6 is a passenger automobile"; and 49 C.F.R. §523.2 defines *light truck* as "a non-passenger automobile meeting the criteria in § 523.5. The term light truck includes medium-duty passenger vehicles that meet the criteria in § 523.5 for non-passenger automobiles."

for each of these different compliance categories typically influences automotive manufacturers' product planning. If NHTSA determines that a manufacturer does not meet the CAFE standard for any of its respective fleets, the manufacturer may face civil penalties. In the 2026 final rule, NHTSA announced amendments to definitions for light-duty vehicle fleet classifications starting in model year 2030. These amendments alter the dividing line between passenger and non-passenger vehicles by including some light trucks that are "primarily designed to move people" in the passenger vehicle fleet classification.¹³ In the 2026 CAFE final rule, NHTSA stated that

separate standards for the passenger car and light truck fleets (referred to by law as passenger automobiles and non-passenger automobiles, and together known as light-duty vehicles) have led manufacturers to reshape the market in unanticipated ways—such as by almost eliminating the production of station wagons (passenger cars that generally have more robust cargo capacity, adding mass and reducing fuel economy) in favor of vehicles like minivans and crossover utility vehicles (considered light trucks, and subject to less stringent standards).¹⁴

U.S. Environmental Protection Agency Vehicle Emissions Rules

Certain thresholds for vehicle performance are set by National Highway Traffic Safety Administration's (NHTSA's) Corporate Average Fuel Economy (CAFE) standards and the Environmental Protection Agency's (EPA's) greenhouse gas (GHG) emission standards.¹⁵ Fuel economy and GHG emissions standards both relate to vehicle fuel consumption. NHTSA and EPA are statutorily required to consider each other's respective standards and consult each other in the rulemaking process.¹⁶ There are also three instances of joint rulemakings on these standards. NHTSA and EPA harmonized greenhouse gas (GHG) emissions and CAFE standards in a 2010 final rule and updated these standards in a 2012 final rule, which was referred to as the "National Program."¹⁷ EPA and NHTSA issued a joint rulemaking in 2020 under "One National Program," which was a midterm evaluation of the 2010 and 2012 final rules.¹⁸ Differing statutory authorities for these standards, changes to the compliance regime, varying levels of industry input, and rules issued separately by each agency contribute to industry stakeholders' criticism of what they perceive as a lack of regulatory alignment and undue regulatory compliance burden.¹⁹ In February 2026, EPA rescinded the 2009 endangerment finding, which required EPA to issue GHG emission standards for vehicles and was a prerequisite for the agency to regulate mobile source GHG emissions.²⁰ Absent GHG emissions standards for vehicles set by the EPA, there are no GHG standards to which NHTSA would align CAFE standards, which would make this regulatory alignment inapplicable.

Statutory Obligations for Standard Setting

Title 49, Section 32902, of the *U.S. Code* sets several statutory obligations for CAFE standards, including certain considerations when determining their stringency. This section of the *U.S. Code*

¹³ NHTSA, "SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks," 91 *Federal Register* 61988, 61997.

¹⁴ NHTSA, "SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks," 91 *Federal Register* 61988, 61995- 61996.

¹⁵ 49 U.S.C. §39202(a); and 42 U.S.C. §7521(a).

¹⁶ 42 U.S.C. §7590(d); 49 U.S.C. §39202(b); and 49 U.S.C. §39202(f).

¹⁷ EPA and NHTSA, "Light-Duty Vehicle Greenhouse Gas Emission Standards and Corporate Average Fuel Economy Standards; Final Rule," 75 *Federal Register* 25324, May 7, 2010; and EPA and NHTSA, "2017 and Later Model Year Light-Duty Vehicle Greenhouse Gas Emissions and Corporate Average Fuel Economy Standards," 77 *Federal Register* 62624, October 15, 2012.

¹⁸ EPA and NHTSA, "The SAFE Vehicles Rule for Model Years 2021-2026 Passenger Cars and Light Trucks," 85 *Federal Register* 24174, April 30, 2020.

¹⁹ George Weykamp, "New CAFE mpg Standards to Take a Back Seat to EPA Rules," *Automotive News*, June 13, 2024, <https://www.autonews.com/regulation-safety/new-cafe-fuel-economy-standards-take-back-seat-epa-rules/>.

²⁰ EPA, "Rescission of the Greenhouse Gas Endangerment Finding and Motor Vehicle Greenhouse Gas Emission (continued...)"

requires the Secretary of Transportation (the Secretary) to prescribe average fuel economy standards at least 18 months before a given model year begins and limits regulation to cover not more than 5 model years.²¹ Title 49, Section 32902(a), requires the Secretary to issue standards that “shall be the maximum feasible average fuel economy level that the Secretary decides the manufacturers can achieve in that model year.”

Congress, via EPCA, required the Secretary to factor in several considerations when assessing the maximum feasible average fuel economy and determining the stringency of CAFE standards.²² These considerations (found in 49 U.S.C. §32902(f)) include “technological feasibility, economic practicability, the effect of other motor vehicle standards of the Government on fuel economy, and the need of the United States to conserve energy.” The following is a brief overview of how NHTSA views these considerations:

- *Technological Considerations*: NHTSA generally does not require specific technologies to meet these standards and must consider the costs and feasibility of technologies available to manufacturers.²³
- *Economic Practicability*: When determining economic practicability and whether the net benefit of the proposed standards exceeding the net costs, NHTSA considers whether the proposed standards are “within the financial capability of the industry, but not so stringent as to threaten substantial economic hardship for the industry.”²⁴
- *Effects of Other Motor Vehicle Emissions and Safety Standards*: NHTSA must consider the effect of “other motor vehicle standards,” including the relationship between CAFE standards and other regulations, such as those relating to safety and emissions.²⁵
- *Energy Conservation*: Energy conservation is interpreted by NHTSA as “the consumer cost, national balance of payments, environmental, and foreign policy implications of our need for large quantities of petroleum, especially imported petroleum.”²⁶

These four factors must always be considered, but Administrations may weigh factors differently depending on their priorities. For example, the Biden Administration set more stringent standards than the Trump Administration and cited energy security (an element of energy conservation) and saving consumers money (an element of economic practicability) as the reasons to set those standards.²⁷ The Trump Administration has set standards that are less stringent and noted the

Standards Under the Clean Air Act,” 91 *Federal Register* 7686, February 18, 2026. For more on the endangerment finding, see CRS Report R49024, *Clean Air Act: EPA’s Greenhouse Gas Endangerment Finding and Repeal*, by Jonathan D. Haskett and Omar M. Hammad.

²¹ 49 U.S.C. §32902(a); and 49 U.S.C. §32902(a)(3)(B).

²² See §502 of P.L. 94-163.

²³ 49 U.S.C. §32902(f).

²⁴ *Public Citizen v. National Highway Traffic Safety Administration*, 848 F.2d 256, 264 (D.C. Cir. 1988); and NHTSA, “SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks,” 90 *Federal Register* 56438, 56585, December 5, 2025.

²⁵ 49 U.S.C. §32902(f).

²⁶ NHTSA, “Passenger Automobile Average Fuel Economy Standards,” 42 *Federal Register* 33534, June 30, 1977; NHTSA, “Non-passenger Automobile Average Fuel Economy Standards Model Years 1980-1981,” 42 *Federal Register* 63188, December 15, 1977; and NHTSA, “SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks,” 91 *Federal Register* 61988, 62189.

²⁷ NHTSA, “CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel (continued...) ”

ability for manufacturers to comply with these standards (elements of technological feasibility and economic practicability) as reasons for more gradual increases in stringency.²⁸

Stringency of CAFE Standards

Congress and DOT have amended and generally increased the stringency of CAFE standards since their implementation in 1978.²⁹ CAFE standards require automotive manufacturers to meet the fuel economy set by the Secretary or otherwise required by Congress.³⁰

EPCA required CAFE standards for both domestic and imported passenger vehicles for model year 1978.³¹ For light-duty trucks, EPCA required the fuel economy to be set for model year 1979; these standards have been raised at points by both NHTSA and Congress since then.³²

The Energy Independence and Security Act of 2007 (EISA; P.L. 110-140) required the Secretary to issue CAFE standards that incrementally increase over time, reaching 35 mpg for the “combined fleet” (or the average of passenger cars and light trucks) by 2020. NHTSA issued its first combined fleet standards in 2009 with a target of 27.3 mpg by model year 2011.³³ Separate standards apply for each of the three light-duty compliance categories. Manufacturers are to achieve standards separately across these compliance categories in addition to the combined fleet standard. For example, regulations issued during the Biden Administration projected a required average fuel economy of 50.4 mpg by model year 2031 for light-duty vehicles.³⁴ In the 2026 CAFE final rule, which revised these standards, NHTSA projected a required average fuel economy of 34.9 mpg by model year 2031 for light-duty vehicles.³⁵

CAFE Targets

NHTSA expresses CAFE standards through a mathematical function, which may be referred to as the “CAFE curve.” The CAFE curve is used to set CAFE targets for manufacturers. EPCA, as amended by EISA, requires CAFE standards to be “based on 1 or more vehicle attributes related to fuel economy and express each standard in the form of a mathematical function.”³⁶ The mathematical function that NHTSA uses to derive the curve for CAFE standards sets individual

Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond,” 89 *Federal Register* 52540, 52832, June 24, 2024.

²⁸ NHTSA, “SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks,” 91 *Federal Register* 61988.

²⁹ P.L. 94-163.

³⁰ 49 U.S.C. § 32902(a); and 49 U.S.C. § 32902(b)(4).

³¹ P.L. 94-163; and U.S. Energy Information Administration, “Fuel Economy Standards Have Affected Vehicle Efficiency,” press release, August 3, 2012, <https://www.eia.gov/todayinenergy/detail.php?id=7390>.

³² NHTSA, *Summary of Fuel Performance*, December 15, 2014, <https://www.nhtsa.gov/sites/nhtsa.gov/files/performance-summary-report-12152014-v2.pdf>; and NHTSA, “Average Fuel Economy Standards for Light Trucks Model Years 2008-2011,” 71 *Federal Register* 17566, April 6, 2006.

³³ P.L. 110-140; and NHTSA, “Average Fuel Economy Standards Passenger Cars and Light Trucks Model Year 2011,” 74 *Federal Register* 14196, March 30, 2009.

³⁴ NHTSA, “Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond,” 89 *Federal Register* 52540, June 24, 2024.

³⁵ NHTSA, “SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks,” 91 *Federal Register* 61988.

³⁶ 49 U.S.C. 32902(a)(3)(A).

CAFE targets for each vehicle based on a size metric, which is called footprint.³⁷ Generally, the larger the vehicle footprint, the lower the target fuel economy for that vehicle.³⁸ The footprint-based targets for each vehicle are then factored into the average fuel economy calculation for a manufacturer's fleet, so each manufacturer may have different CAFE targets depending on the footprint of the vehicles in their fleet.³⁹

Compliance

NHTSA and the Environmental Protection Agency (EPA) both have roles in the enforcement of CAFE standards. Before and during the model year, manufacturers submit reports to NHTSA; at the end of a model year, manufacturers must submit compliance reports to EPA.⁴⁰ Manufacturers are required to provide EPA with data that include sales figures for the model year. EPA uses this data to determine whether manufacturers met the standards for the CAFE set for their fleet.⁴¹ EPA may verify manufacturers' data, calculate manufacturers' CAFE values, determine the average CAFE values, and issue final CAFE reports to manufacturers and NHTSA. EPA also conducts tests to measure the fuel economy and emissions of the vehicles.⁴² After EPA issues these reports to NHTSA, NHTSA identifies whether the fleet performance is greater than or equal to the standards set for the manufacturer and determines credit balances for the manufacturer.⁴³ NHTSA is to notify a manufacturer if it fails to meet the CAFE standard for its fleet or offer credits if it overcomplies with its CAFE standards.

Credits

NHTSA offers credits for overcompliance if a manufacturer exceeds its CAFE standard for a fleet.⁴⁴ A manufacturer may earn one credit for every "1/10 of a mile per gallon above the fuel economy standard per one vehicle within a compliance category."⁴⁵ A credit can be applied to a previous model year up to three years before it was earned (which is sometimes referred to as a "carryback"), applied to future model years up to five model years after it was earned (sometimes referred to as a "carryforward"), transferred to another fleet of the manufacturer, or traded to another manufacturer.⁴⁶ Carrybacks give flexibilities to manufacturers so that if they miss their target for a given model year, they can make up for the shortfall by exceeding standards in the

³⁷ In 49 C.F.R. §523.2, the term *footprint* is defined as the product of track width (calculated as the average of the front and rear track widths) and wheelbase (rounded to the nearest tenth of a square foot).

³⁸ NHTSA, "Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond," 89 *Federal Register* 52540, 52567, June 24, 2024.

³⁹ NHTSA, "Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond," 89 *Federal Register* 52540, 52567, June 24, 2024.

⁴⁰ NHTSA, "CAFE Public Information Center," accessed April 7, 2026, <https://www.nhtsa.gov/corporate-average-fuel-economy/cafe-public-information-center>.

⁴¹ 40 C.F.R. Part 600, Subpart F; and 49 C.F.R. §537.9(b).

⁴² For more on EPA test procedures, see 40 C.F.R. §600; 49 U.S.C. §32907; and EPA, *The 2025 EPA Automotive Trends Report*, February 2026, <https://www.epa.gov/system/files/documents/2026-02/420r26001.pdf>.

⁴³ NHTSA, "CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond," 89 *Federal Register* 52540, 52919, June 24, 2024.

⁴⁴ A manufacturer's ability to earn credits was authorized by P.L. 103-272.

⁴⁵ 49 C.F.R. §536.3(b)(7).

⁴⁶ 49 U.S.C. §32912.

upcoming model years and apply those credits earned to a year when they missed the target and have not yet paid a penalty. Carryforwards offer similar flexibility for future model years. In 2007, Congress amended Title 49, Section 32903, of the *U.S. Code* through EISA to increase the carryforward limit from three to five years.⁴⁷

Manufacturers can also transfer credits earned in one fleet (i.e., domestic passenger vehicles, imported passenger vehicles, and light trucks) to another fleet.⁴⁸ For example, if a manufacturer overcomplies with its standard for its imported passenger vehicles but falls short for light trucks, the manufacturer can apply the credits earned from imported passenger vehicles to light trucks.

“Inter-manufacturer” credit trading, authorized by EISA in 2007, allows manufacturers to trade credits, in exchange for money (referred to as selling credits), with other manufacturers.⁴⁹

NHTSA is statutorily prohibited from considering credit trading and transferring when setting standards.⁵⁰ Some manufacturers, such as those that manufacture alternative fuel vehicles or only electric vehicles (EVs), generally overcomply with their CAFE standards and generate more credits than other manufacturers. Some of these manufacturers have regularly sold credits to other manufacturers, as part of a market for credit trading.⁵¹ If a manufacturer chooses to trade credits to another manufacturer, both manufacturers must report the trade to NHTSA and provide certain required information.⁵²

Since inter-manufacturer credit trading was introduced, a multibillion dollar credit trading market has emerged. Some manufacturers that primarily produce alternative fuel vehicles reportedly have particularly benefitted from this market by generating and trading credits to manufacturers that did not meet their CAFE standards.⁵³ In the 2026 CAFE final rule, NHTSA announced the removal of inter-manufacturer credit trading from the CAFE program beginning with credits earned in model year 2028, citing the advantages for EV manufacturers and the capital constraints arising from purchasing credits for other manufacturers.⁵⁴

Civil Penalty

If a manufacturer fails to meet CAFE standards and does not utilize credits to comply, it faces a civil penalty.⁵⁵ In some cases, manufacturers may fail to meet a CAFE standard for reasons they might not have projected. For example, when consumers buy more large (and generally less fuel-efficient) vehicles from a fleet than the manufacturer anticipates, it may fail to meet the CAFE standard set for its fleet.⁵⁶ Consequently, automotive manufacturers may choose to compensate

⁴⁷ P.L. 110-140, §104.

⁴⁸ 49 C.F.R. §536.3.

⁴⁹ P.L. 110-140, §104.

⁵⁰ 49 U.S.C. §32902(h).

⁵¹ Benjamin Leard and Virginia McConnell, *New Markets for Credit Trading under US Automobile Greenhouse Gas and Fuel Economy Standards*, Resources for the Future, August 2017, <https://media.rff.org/documents/RFF-Rpt-AutoCreditTradingREV.pdf>.

⁵² 49 C.F.R. §536.8(a).

⁵³ Ryan Felton and Sharon Terlep, “Rivian Says It Faces \$100 Million Hole After Relaxation of Fuel Economy Rules,” *Wall Street Journal*, August 14, 2025, <https://www.wsj.com/business/autos/rivian-ev-fuel-economy-rules-36ebe6f4>.

⁵⁴ NHTSA, “SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks,” 91 *Federal Register* 61988, 62259.

⁵⁵ 49 U.S.C. §32912.

⁵⁶ For civil penalties paid by automotive manufacturers, see NHTSA, “CAFE Public Information Center: Summary of CAFE Civil Penalties Collected,” updated July 23, 2025, <https://explore.dot.gov/t/NHTSA-CAFE/views/> (continued...)

for this by offering discounts in the given model for more fuel-efficient vehicles, producing more fuel-efficient vehicles in the following model year, purchasing credits, or paying civil penalties.

Since the inception of the CAFE program, manufacturers have regularly paid civil penalties (see **Figure 1**). EPCA originally set civil penalties at \$5.00 “for each tenth of a mile per gallon by which the average fuel economy of the passenger automobiles manufactured by such manufacturer during such model year is exceeded by the applicable average fuel economy standard established.”⁵⁷ The civil penalty was raised to \$5.50 in 1997, and a rule was finalized in 2016 to raise it to \$14.00. This increase was delayed by the Trump Administration, reinstated by the Biden Administration, and then raised by the Biden Administration to \$15.00 for model year 2022, \$16 for model year 2023, and \$17 for model year 2024.⁵⁸ Section 40006 of the FY2025 reconciliation law (P.L. 119-21) statutorily reduced civil penalties for the violation of CAFE standards to \$0.00.

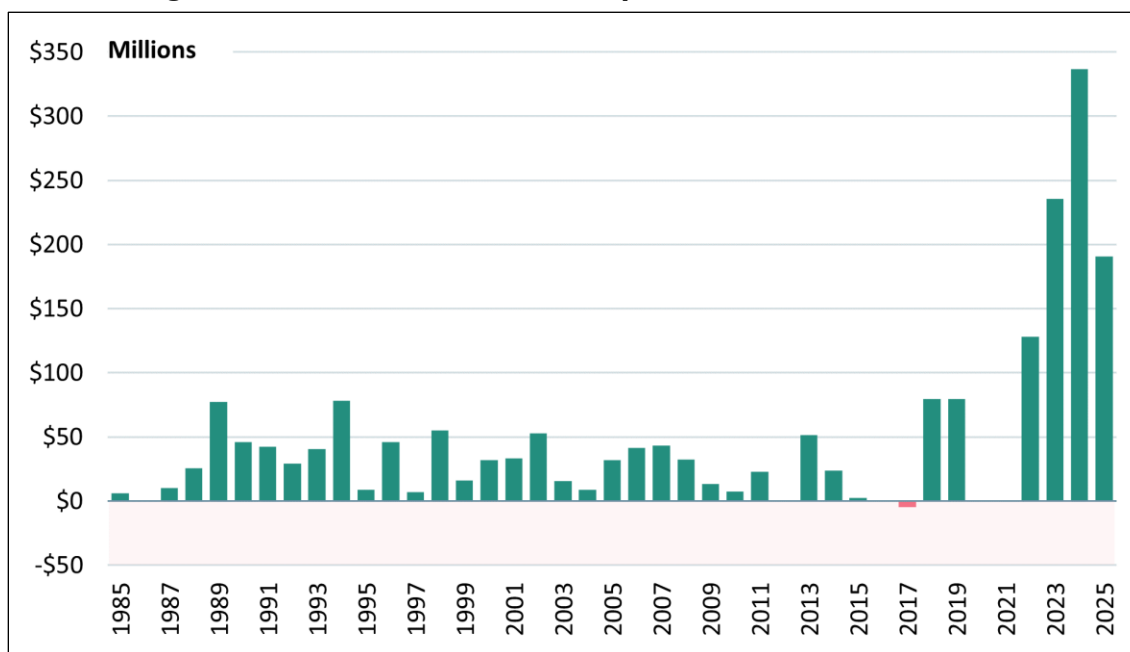
This reduction in the civil penalty to \$0.00 may lessen the incentives for manufacturers to produce fuel-efficient vehicles and technologies that support fuel economy improvements. Since manufacturers have factored the payment of penalties as a potential business cost over time, the civil penalty reduction to \$0.00 may affect their future business decisions.⁵⁹ Additionally, a civil penalty set at \$0.00 may remove the market for manufacturers to purchase credits from each other to avoid civil penalties. This may have differential effects on the automobile manufacturing industry depending on the extent to which a company was the purchaser or provider of credits within the fuel economy credit market.

CAFEPICTReport-CivilPenalties/SummaryofCAFECivilPenaltiesCollectedDashboard?%3Adisplay_count=n&%3Aembed=y&%3AisGuestRedirectFromVizportal=y&%3Aorigin=viz_share_link&%3AshowAppBanner=false&%3AshowVizHome=n.

⁵⁷ P.L. 94-163, §508.

⁵⁸ NHTSA, “Civil Penalties,” 87 *Federal Register* 18994, April 1, 2022; and the U.S. Department of Transportation, “Revisions to Civil Penalty Amounts, 2024,” 88 *Federal Register* 89551, December 18, 2023.

⁵⁹ DOT, “CAFE Public Information Center: Summary of Civil Penalties Collected,” updated July 23, 2025, https://explore.dot.gov/t/NHTSA-CAFE/views/CAFEPICReport-CivilPenalties/SummaryofCAFECivilPenaltiesCollectedDashboard?%3Adisplay_count=n&%3Aembed=y&%3AisGuestRedirectFromVizportal=y&%3Aorigin=viz_share_link&%3AshowAppBanner=false&%3AshowVizHome=n.

Figure I. Civil Penalties Collected by NHTSA from 1985 to 2025

Source: Adapted by CRS from U.S. Department of Transportation, “CAFE Public Information Center: Summary of Civil Penalties Collected,” updated July 23, 2025, https://explore.dot.gov/t/NHTSA-CAFE/views/CAFEPICReport-CivilPenalties/SummaryofCAFECivilPenaltiesCollectedDashboard?%3Adisplay_count=n&%3Aembed=y&%3AisGuestRedirectFromVizportal=y&%3Aorigin=viz_share_link&%3AshowAppBanner=false&%3AshowVizHome=n.

Note: Negative civil penalties collected mean that the agency reimbursed more money from penalties than it collected.

Labeling

Under Title 49, Section 32908, of the *U.S. Code*, automotive manufacturers are required to label certain fuel economy information on vehicles. These labeling requirements are administered through coordination between EPA and NHTSA, with EPA as the agency primarily responsible for defining test procedures, testing vehicle fuel economy, and confirming some vehicle test results, which in part determine the values provided on the fuel economy label.⁶⁰ These labels were first required in the 1970s by EPCA and have since changed.⁶¹ These changes include the incorporation of a projected five-year fuel cost saving compared with the average new vehicle, driving range and charge time for EVs, and ratings on a vehicle’s smog and GHG emissions.⁶²

⁶⁰ EPA, “Fuel Economy and EV Range Testing,” accessed July 18, 2025, <https://www.epa.gov/greenvehicles/fuel-economy-and-ev-range-testing>.

⁶¹ P.L. 94-163; and EPA, “History of Fuel Economy Labeling,” updated May 21, 2025, <https://www.epa.gov/fueleconomy/history-fuel-economy-labeling>.

⁶² EPA, “Learn About the Fuel Economy Label,” updated December 5, 2025, <https://www.epa.gov/greenvehicles/learn-about-fuel-economy-label>.

Fuel Consumption Improvement Values (FCIVs)

The CAFE program offers manufacturers flexibilities in the form of FCIVs, which require manufacturers to demonstrate certain fuel efficiency improvements through technologies. These FCIVs can take the form of improvements in air conditioning (AC) efficiency technologies and off-cycle (OC) technologies (e.g., thermal control technologies, high-efficiency alternators, high-efficiency exterior lighting, and engine idle start-stop technologies).⁶³ These flexibilities were introduced to the CAFE program in 2012 when NHTSA issued a joint rulemaking with EPA on CAFE and EPA GHG standards. EPA already recognized FCIVs at this time, and the joint final rule integrated these technologies into CAFE compliance.⁶⁴ AC and OC technologies are factored into EPA's calculations when determining CAFE compliance and can be used to improve the average fuel economy of a manufacturer's overall fleet.⁶⁵ NHTSA's 2024 CAFE final rule phases out OC FCIVs by model year 2033 to align with EPA's phasing out of OC FCIVs for GHG emissions standards.⁶⁶ In the 2026 CAFE final rule, the agency announced the removal of OC and AC FCIVs from standard-setting analysis beginning in 2028 to ensure that projected standards are achievable for manufacturers without these technologies; NHTSA also announced the removal of references to EPA's AC and OC regulations.⁶⁷

Alternative Fuel Vehicles

Alternative fuel vehicles include electric, ethanol, biodiesel, and natural gas vehicles. They are incorporated into CAFE standards through flexibilities and limitations. Alternative fuel vehicles in CAFE standards are inclusive of dual-fueled and dedicated fueled vehicles.⁶⁸ Dual-fueled vehicles could refer to plug-in hybrid vehicles or flexible-fuel vehicles (FFVs).⁶⁹ Alternative fuel vehicles were first granted an incentive in CAFE calculations by the Alternative Motor Fuels Act of 1988 (AMFA; P.L. 100-494).⁷⁰

For dual-fueled vehicles, regulators constructed the fuel economy formula with the assumption that half of the fuel utilized was gasoline and the other half was alternative fuel, which was given a 0.15 divisor as an incentive.⁷¹ The assumption that these dual-fueled vehicles were operating on half gasoline and half alternative fuel received criticism from environmental advocacy

⁶³ 49 C.F.R. §533.6(c); 40 C.F.R. §86.1868-12; and 40 C.F.R. §86.1869-12(b).

⁶⁴ EPA and NHTSA, "2017 and Later Model Year Light-Duty Vehicle Greenhouse Gas Emissions and Corporate Average Fuel Economy Standards," 77 *Federal Register* 62624, October 15, 2012.

⁶⁵ 40 C.F.R. §600.510-12.

⁶⁶ NHTSA, "CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond," 89 *Federal Register* 52540, 52929, June 24, 2024.

⁶⁷ NHTSA, "SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks," 91 *Federal Register* 61988, 62005.

⁶⁸ 40 C.F.R. §600.002 defines a *dual fueled automobile or vehicle* as a vehicle "which is designed to operate on an alternative fuel and on gasoline or diesel fuel." A *dedicated* fueled vehicle is defined at 49 U.S.C. §32901(a)(8) as "an automobile that operates only on alternative fuel." Dedicated fueled vehicles could include a vehicle that strictly runs on an electric battery, compressed natural gas, or biodiesel.

⁶⁹ Flexible-fuel vehicles (FFVs) are defined at 40 C.F.R. §86.1803-01 as "any motor vehicle engineered and designed to be operated on a petroleum fuel and on a methanol or ethanol fuel, or any mixture of the petroleum fuel and methanol or ethanol." FFVs could be capable of running on gasoline and ethanol.

⁷⁰ Gregory Powell, *Alternative Fuels in CAFE Rulemaking*, NHTSA, January 2015, https://www.nhtsa.gov/sites/nhtsa.gov/files/2015sae-powell-altfuels_cafe.pdf.

⁷¹ 49 U.S.C. §32905(b)(1)(2).

stakeholders in the case of FFVs. The incentive for FFVs received some criticism because of instances where these vehicles used less than the assumed amount of ethanol.⁷² The maximum increase in average fuel economy that nonelectric dual-fueled vehicles (e.g., FFVs) could receive from this incentive was originally set at 1.2 mpg in 1993. In 2007, EISA decreased the maximum increase in average fuel economy for nonelectric dual-fueled vehicles from this incentive to 1.0 mpg by model year 2015, decreasing 0.2 mpg each model year until it reached 0 mpg after model 2019.⁷³

EISA required EPA to update these calculations without these incentives after 2019.⁷⁴ This resulted in calculations that weighed the utilization of alternative fuels instead of the assumption of 50/50 gasoline and alternative fuel.⁷⁵ Additionally, the National Defense Authorization Act for Fiscal Year 2015 (P.L. 113-291) required EPA to measure the fuel economy of electric dual-fueled vehicles using a similar weighting of the vehicle's utilization of electricity and gasoline or diesel.⁷⁶ Some alternative energy producers and agricultural stakeholders requested that NHTSA and EPA reinstate the prior incentives for FFVs in the 2024 CAFE rulemaking.⁷⁷ NHTSA did not propose further incentives but continued to allow FFVs to receive credits if they overcomplied with standards.⁷⁸

Specific calculations for EVs, such as battery-electric and plug-in hybrid, were not formalized until 2000. This is done by calculating a petroleum equivalency factor (PEF) that allows automotive manufacturers to convert the measured electrical energy consumption of EVs into an mpg equivalent and factor it into the CAFE standards of a fleet.⁷⁹ In 2000, and updated in 2024, DOE published a final rule with procedures for calculating the PEF of EVs.⁸⁰ In February 2026, DOE issued an interim final rule revising the PEF calculation by removing a consideration called the fuel content factor (FCF) from the calculation. DOE cited that this removal is “consistent with a United States Court of Appeals for the Eighth Circuit decision that held, among other things, that the inclusion of the FCF in the PEF calculation exceeded DOE’s authority under the substantive statute.”⁸¹

⁷² National Public Radio, “‘Flex-Fuel’ Concept Fails to Deliver on Potential,” December 26, 2005, <https://www.npr.org/2005/12/26/5070205/flex-fuel-concept-fails-to-deliver-on-potential>.

⁷³ 49 U.S.C. §32906(a).

⁷⁴ P.L. 110-140.

⁷⁵ 49 U.S.C. §32906(b); 40 C.F.R. §600.510-12(c)(2); and EPA, “Technical Correction to the Flex-Fuel Vehicle Provisions in CAFE Regulations,” 85 *Federal Register* 53676, August 31, 2020.

⁷⁶ 49 U.S.C. §32905(e).

⁷⁷ For example, see Comments on Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027–2032 and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030–2035 from the Renewable Fuels Association, National Corn Growers Association, and National Farmers Union to NHTSA, August 17, 2023.

⁷⁸ NHTSA, “CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond,” 89 *Federal Register* 52540, 52936, June 24, 2024.

⁷⁹ For more on the history and calculation of the petroleum-equivalent fuel economy of electric vehicles, see CRS Report R48086, *Petroleum-Equivalent Fuel Economy of Electric Vehicles: In Brief*, by Corrie E. Clark and Richard K. Lattanzio.

⁸⁰ DOE, “Electric and Hybrid Vehicle Research, Development, and Demonstration Program; Petroleum-Equivalent Fuel Economy Calculation,” 65 *Federal Register* 36986, June 12, 2000; and DOE, “Petroleum-Equivalent Fuel Economy Calculation,” 89 *Federal Register* 22041, March 9, 2024.

⁸¹ DOE, “Petroleum-Equivalent Fuel Economy Calculation,” 91 *Federal Register* 7810, February 19, 2026; and *Iowa v. Wright*, 154 F.4th 918, 946 (8th Cir. 2025).

NHTSA is statutorily required to consider dual-fueled vehicles as operated only on gasoline or diesel fuel and may not consider the fuel economy of dedicated automobiles when setting new CAFE standards.⁸² NHTSA may take other approaches when considering alternative fuel vehicles during the rulemaking process. In 2024, NHTSA included alternative fuel vehicles in other aspects of the standards, such as including EVs when considering a no-action alternative, commonly referred to as the “reference baseline.”⁸³ Some automotive industry stakeholders claimed the inclusion of EVs in the analysis of the reference baseline was statutorily prohibited by EPCA and EISA. Specifically, these stakeholders argued that NHTSA is statutorily prohibited from considering the fuel economy of dedicated alternative fuels in its consideration of CAFE standards.⁸⁴ NHTSA determined that this is only the case for the analysis of what is required to meet NHTSA’s final CAFE standards, not the assumptions in the reference baseline.⁸⁵ Additionally, NHTSA noted that “modeling ... electric vehicles that manufacturers have committed to deploy enables NHTSA to make more realistic projections of how the U.S. vehicle fleet will change in the coming years independent of CAFE standards, which is foundational to our ability to set CAFE standards that reflect the maximum feasible fuel economy level achievable through improvements to internal combustion vehicles.”⁸⁶ In the 2026 final rule, NHTSA stated that it did not consider EVs when setting standards and noted a differing approach than CAFE rulemakings from 2020, 2022, and 2024.⁸⁷

Selected Policy Options

Congress established the CAFE program through EPCA and has amended it through legislation such as EISA. NHTSA generally administers this program, and Congress may choose to reevaluate the program and agency authorities. Lawmakers may choose to evaluate the program and engage in oversight through hearings, requests for information, or consultations with NHTSA and industry stakeholders through meetings or advisory programs.

Congress may choose to reevaluate some aspects of the CAFE program, such as vehicle fleet compliance categories, statutory obligations, the stringency of the standards, harmonization with EPA GHG emissions standards, enforcement, and the various flexibilities offered by the program. Congress may consider whether, and if so how, these aspects might further the program’s goals, such as energy conservation; a consumer’s ability to save on fuel costs; the industry’s ability to produce vehicles that comply with proposed standards; and the relationship between CAFE standards and vehicle safety or fuel consumption generally. Members may assess whether, and, if

⁸² 49 U.S.C. §32902(h)(2); and 49 U.S.C. §32902(h)(1).

⁸³ NHTSA, “CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond,” 89 *Federal Register* 52540, 52785-52786, June 24, 2024.

⁸⁴ For example, see Comments on Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027–2032 and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030–2035 from the Alliance for Automotive Innovation to NHTSA, October 16, 2023; NHTSA, “CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond,” 89 *Federal Register* 52540, 52785-52786, June 24, 2024.

⁸⁵ NHTSA, “CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond,” 89 *Federal Register* 52540, 52785-52786, June 24, 2024.

⁸⁶ NHTSA, “CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond,” 89 *Federal Register* 52540, 52895, June 24, 2024.

⁸⁷ NHTSA, “SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks,” 91 *Federal Register* 61988, 62203-62204.

so, to what extent, the reduction of the civil penalty for violating CAFE standards has affected the incentives for manufacturers to comply with the regulation. The following are selected potential considerations for Congress.

Vehicle Fleet Definitions

Members may examine vehicle fleet classifications. Title 49, Section 32902, of the *U.S. Code* provides requirements to set separate standards for vehicle fleets, and Title 49, Section 523, of the *Code of Federal Regulations* sets out definitions of these different types of vehicle fleets. The definitions of *domestic passenger vehicles*, *imported passenger vehicles*, and *light trucks* and their associated requirements were amended in the 2026 CAFE final rule. The 2026 CAFE final rule adjusted the passenger vehicle fleet to consist of “vehicles primarily designed to move people” and the light truck fleet to consist of “vehicles primarily designed to operate off highway or move cargo.”⁸⁸ Congress may consider the definitions of light-duty fleet classifications to account for changing vehicle uses, such as light trucks being used for passenger purposes, or vehicle fleet classifications incentivizing vehicle manufacturing misaligned with consumers’ interests.

Standard Setting

Congressional policymakers may choose to review aspects of Title 49, Section 32902, of the *U.S. Code*, which requires the Secretary of Transportation to factor in several considerations when determining the maximum feasible average fuel economy. These considerations include “technological feasibility, economic practicability, the effect of other motor vehicle standards of the Government on fuel economy, and the need of the United States to conserve energy” (see “Statutory Obligations for Standard Setting”).⁸⁹ Administrations may weigh these considerations differently, which might result in CAFE standard changes based on such differing assessments. Automotive industry companies may find complying with changing standards costly or burdensome. For example, the automotive industry typically develops 5- to 10-year product plans, so CAFE standards that change from one Administration to another might affect or disrupt such product plans. Congressional policymakers may consider whether potential regulatory uncertainties support further definition to reduce variability in standard setting.⁹⁰

Policymakers might similarly assess the stringency of CAFE standards and the extent to which such stringency meets the intent of Congress. Congress could itself establish CAFE targets. For example, lawmakers set the target of 35 mpg by model year 2020 for light-duty vehicles when it enacted EISA in 2007.⁹¹ Since then, NHTSA has set various CAFE standards through rulemakings over multiple Administrations. Alternatively, Congress might create targets for individual fleets.

Members might weigh the value of providing agency discretion in setting standards versus establishing prescriptive standards in statute. Continuing, or refining, agency discretion may allow for more timely response to technological advances, while establishing prescriptive standards in statute may serve as a baseline or indicate the intent of Congress with respect to stringency or the trajectory of the standard.

⁸⁸ NHTSA, “SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks,” 91 *Federal Register* 61988, 61997.

⁸⁹ See P.L. 94-163, §502; and 49 U.S.C. §32902(f).

⁹⁰ 49 U.S.C. §32902(f).

⁹¹ P.L. 110-140.

Role of EPA

Both EPA and NHTSA have roles regarding the regulation vehicle performance. Policymakers may consider the alignment of NHTSA CAFE standards with EPA GHG emissions standards efforts or EPA's involvement in verifying manufacturer compliance with CAFE standards. NHTSA has considered EPA GHG emissions standards in all its CAFE rulemakings, and the agencies have issued joint rulemakings (to varying degrees) three times under their National Program.⁹² NHTSA and EPA have frequently noted statutory obligations to consider the CAFE and GHG emissions when setting new standards, but some automotive industry stakeholders look for further coordination.⁹³

Some industry stakeholders have called for alignment between NHTSA and EPA standards, reasoning that compliance for fuel consumption regulations may be more complex when not aligned across agencies,⁹⁴ while the rescission of the 2009 endangerment finding may lessen calls for harmonization between NHTSA and EPA standards.⁹⁵ In previous rulemakings, the agencies have cited their different statutory authorities as limitations to fully aligning their standards.⁹⁶ Members could consider addressing statutory authorities that may affect standard setting, required metrics for determining compliance, and timelines for rulemakings. Congress could determine whether to require the agencies to issue joint rulemakings, for example, as was done in the National Program rulemaking, in an effort to ensure alignment.

EPA also oversees vehicle fuel economy testing and generates manufacturers' CAFE reports to verify compliance; interested Members might choose to review this EPA function. In 2005, Congress required EPA to update aspects of fuel economy testing procedures to ensure they were more reflective of actual in-use fuel economy of vehicles.⁹⁷ Interested Members might continue oversight regarding testing procedures and the alignment of these results with vehicle label information regarding fuel economy.

⁹² EPA and NHTSA, "Light-Duty Vehicle Greenhouse Gas Emission Standards and CAFE Standards; Final Rule," 75 *Federal Register* 25324, May 7, 2010; EPA and NHTSA, "2017 and Later Model Year Light-Duty Vehicle Greenhouse Gas Emissions and Corporate Average Fuel Economy Standards," 77 *Federal Register* 62624, October 15, 2012; and EPA and NHTSA, "The Safer Affordable Fuel-Efficient (SAFE) Vehicles Rule for Model Years 2021-2026 Passenger Cars and Light Trucks," 85 *Federal Register* 24174, April 30, 2020.

⁹³ NHTSA, "CAFE Standards for Model Years 2024-2026 Passenger Cars and Light Trucks," 87 *Federal Register* 25710, May 2, 2022; NHTSA, "CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond," 89 *Federal Register* 52540, June 24, 2024; and for an example of an industry stakeholders perspective, see Comments on Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027–2032 and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030–2035 from Joanna Foust, vice president, Government and Regulatory Affairs, North American Subaru, Inc. to NHTSA, p. 2, October 16, 2023, and Comments on Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027–2032 and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030–2035 from the Alliance for Automotive Innovation to NHTSA, October 16, 2023.

⁹⁴ Weykamp, "New CAFE mpg Standards to Take a Back Seat to EPA Rules."

⁹⁵ EPA, "Rescission of the Greenhouse Gas Endangerment Finding and Motor Vehicle Greenhouse Gas Emission Standards Under the Clean Air Act," 91 *Federal Register* 7686, February 18, 2026.

⁹⁶ NHTSA, "CAFE Standards for Model Years 2024-2026 Passenger Cars and Light Trucks," 87 *Federal Register* 25710, May 2, 2022; and NHTSA, "CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond," 89 *Federal Register* 52540, June 24, 2024.

⁹⁷ See P.L. 109-58, §774.

Compliance Enforcement

In the FY2025 reconciliation law, Congress set the civil penalty for not complying with CAFE standards to \$0.00 per tenth mpg.⁹⁸ The civil penalty is NHTSA's primary mechanism for addressing noncompliance with CAFE standards and thus incentivizing manufacturers to produce more fuel-efficient vehicles and support the goals of the program.

Members could evaluate the relationship between NHTSA and automotive manufacturers with respect to incentivizing production of fuel-efficient vehicles and what alternative mechanisms exist to achieve compliance. For example, these alternative mechanism could include directly engaging with the industry to align on plans to produce more fuel-efficient fleets, similar to the National Program, or creating grant programs and tax credits to support the production and sale of fuel-efficient vehicle, similar to programs found in the Infrastructure Investment and Jobs Act (P.L. 117-58) and the FY2022 budget reconciliation measure (P.L. 117-169).⁹⁹

Members might engage in oversight activities to determine the effect on fleet fuel efficiency of reducing the civil penalty. Absent the presence of a civil penalty, fleet fuel efficiency might decrease and the broad goals of the statute might not be met. In contrast, market forces or technology development may be sufficient to incentivize manufacturers to meet current or exceed current fuel efficiency standards. Such information might inform decisions regarding the sufficiency of the existing penalty, the apparent extent of vehicle fuel efficiency, and potential alternative mechanisms that may exist to achieve compliance.

Flexibilities

NHTSA offers several flexibilities to help manufacturers comply with CAFE standards and account for alternative fuel vehicles. These include credits for overcompliance with CAFE standards, FCIVs derived from OC and AC technologies, and incentives in calculations for alternatives fuels. Congress and NHTSA have altered these flexibilities since passage of EPCA, and the automotive industry has responded to these adjustments. Members may choose to evaluate these flexibilities and their applicability to the CAFE program.

Credits

The timeframe for carrying forward and carrying back credits has been amended previously by Congress.¹⁰⁰ Members might explore the effects of extending or reducing the period that manufacturers can carry forward or carry back credits. Such changes to these time periods would determine when manufacturers' credits expire and their potential for use in complying with the program.

The 2026 CAFE final rule announced the removal of inter-manufacturer credit trading beginning with credits earned in model year 2028 and noted that it has "resulted in a windfall for EV-exclusive manufacturers that sell credits to other non-EV manufacturers, which in turn pay for those credits with capital that could be invested toward improving the fuel economy performance

⁹⁸ P.L. 119-21.

⁹⁹ EPA, "Select Federal Tax Credits Under the Infrastructure Investment and Jobs Act and Inflation Reduction Act," updated July 28, 2025, <https://www.epa.gov/inflation-reduction-act/select-federal-tax-credits-under-infrastructure-investment-and-jobs-act-and>; and also, see §§40205-40210 of P.L. 117-58 for relevant provisions, and see §§13401-13404 and §§13501-13502 of P.L. 117-169.

¹⁰⁰ P.L. 110-140, §104.

or other desirable attributes of their traditional fleets.”¹⁰¹ Congress could evaluate this assessment and could consider whether to codify a limitation on inter-manufacturer credit trading. This could potentially adversely affect EV manufacturers, which would lose a source of revenue, and potentially make regulatory compliance more difficult for companies that engage in purchase of these credits. Alternatively, should Congress seek to require the maintenance of inter-manufacturer credit training, it could remove discretionary language provided to the Secretary and replace it with a statutory requirement.¹⁰²

FCIVs

Congress could evaluate whether OC and AC technologies improve vehicle performance and save consumers money that would have been spent on fuel. Members of Congress could assess whether FCIVs lead to these improvements, and could take action by formalizing these technologies in the CAFE program, which may take the form of writing language into the *U.S. Code*, factoring FCIVs into CAFE calculations, or requiring a rulemaking from the agency on these technologies.

Alternatively, Congress could analyze whether these technologies do not improve vehicle performance and do not save consumers more money, and whether these technologies should remain out of consideration for the CAFE program. If this is the case, Congress might allow NHTSA to proceed with the current rulemakings addressing FCIVs in the CAFE program or decide to consider legislation with language limiting the inclusion of OC and AC technologies to ensure that future rulemakings do not factor these technologies back into the program. Lawmakers may also determine that NHTSA is best suited for evaluating these technologies and allow for these decisions on these technologies to be made at the agency’s discretion.

Alternative Fuels

Members may attempt to clarify the role of alternative fuel vehicles (e.g., dedicated and dual-fueled vehicles) in CAFE standards. The inclusion of alternative fuel vehicles in CAFE standards has led to differing policies between Administrations (see “Alternative Fuel Vehicles”). A wide range of stakeholders have expressed diverging opinions, mainly concerning consideration of EVs. Some of these stakeholders have urged NHTSA to remove consideration of EVs in many aspects of CAFE standards;¹⁰³ others have supported NHTSA’s current practices of including these vehicles in standard setting and flexibilities.¹⁰⁴ Congress might evaluate certain types of alternative fuel vehicles separately or jointly when considering their inclusion in CAFE standards as well as the incentives applied to alternative fuel vehicles in calculating CAFE compliance, the credit accrual and trading for these types of vehicles, or these vehicles’ inclusion in standard setting.

¹⁰¹ NHTSA, “SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks,” 91 *Federal Register* 61988, 61996.

¹⁰² As authorized by P.L. 110-140, §104.

¹⁰³ For example, on automotive industry perspectives, in the *Federal Register* summary of comments, commenters that opposed aspects of NHTSA’s approach included the Alliance for Automotive Innovation, BMW, Toyota, Volkswagen, Kia, Stellantis, and the National Automobile Dealers Association. NHTSA, “CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond,” 89 *Federal Register* 52540, 52833, June 24, 2024.

¹⁰⁴ For example, on nongovernmental associations perspectives, see *on Corporate Average Fuel Economy Standards for Passenger Cars and Light Trucks for Model Years 2027–2032 and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030–2035* from Center for Biological Diversity, Natural Resources Defense Council, Sierra Club, and Union of Concerned Scientists to NHTSA, October 16, 2023.

Incentives

Congress has modified the incentives for alternative fuel vehicles through law several times and could choose to revise or introduce further incentives applied to FFVs, plug-in hybrid vehicles, nonelectric dedicated fueled vehicles, and EVs.¹⁰⁵ Congress might reevaluate the calculation for these alternative fuel vehicles and alter incentives in the calculation, depending on lawmakers' assessment of how these alternative fuel vehicles support the goals of the CAFE program, such as energy conservation.¹⁰⁶ If there are low adoption rates of alternative fuel vehicles that would support energy conservation and lawmakers are interested in incentivizing the production of these vehicles, Congress may seek provide incentives in calculations. Alternatively, if lawmakers determine that these vehicles do not further the goals of the CAFE program, Members may opt against providing further incentives or adjust the current incentive structure for alternative fuel vehicles. For example, lawmakers may determine that the current incentive structure is overly favorable to alternative fuel vehicles and decide to adjust the consideration of these vehicles in the CAFE program. If lawmakers decide to adjust these considerations of alternative vehicles in either of the aforementioned ways, they may adjust the incentives these vehicles receive by directing NHTSA to engage in a rulemaking on this topic or adjusting the statutory language. Additionally, Members of Congress may determine that the CAFE program's current way of incorporating alternative fuel vehicles into the program is sufficient and decide to leave it to NHTSA's discretion if there are to be further adjustments to how these vehicles are considered.

Credit Accrual and Trading

Recent rulemaking activity could alter the ability of manufacturers of alternative fuel vehicles to accrue and trade credits. The 2026 CAFE final rule announced the removal of credit trading beginning with credits earned in model year 2028 and noted as a reason for this decision the monetary advantages credit trading gives manufacturers that produce only EVs over automotive manufacturers that produce internal combustion engine vehicles.¹⁰⁷ Some environmental advocacy stakeholders oppose the Trump Administration's removal of credit trading and EVs from all aspects of standard setting and have commented that this change would not support the CAFE program's goal of reducing oil consumption.¹⁰⁸

Congress could maintain the current regulatory framework for credit trading and rely on the expertise of NHTSA for these decisions. Providing such discretion to NHTSA might lead to further changes to credit trading depending on Administration priorities, potentially affecting industry long-term planning.

Lawmakers may pursue legislation on this topic to ensure more regulatory certainty, such as by barring alternative fuel vehicles from accruing and trading credits, affirming their ability to accrue and trade credits, or specifying types of alternative fuel vehicles that can accrue and trade credits. Congress could evaluate whether the current credit system unfairly advantages EV manufacturers, and Members could consider whether to introduce legislation that would prohibit

¹⁰⁵ P.L. 96-185; P.L. 100-494; P.L. 110-140; and P.L. 113-291.

¹⁰⁶ 49 U.S.C. §32905(a); and 49 U.S.C. §32905(g).

¹⁰⁷ NHTSA, "SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks," 91 *Federal Register* 61988, 62259.

¹⁰⁸ Comments on the Proposed Safer Affordable Fuel-Efficient (SAFE) Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks from Natural Resources Defense Council et al., to NHTSA, February 4, 2026, <https://www.nrdc.org/sites/default/files/2026-02/nrdc-et-al-comments-on-nhtsa-proposed-standards-for-2022-to-2031-comment.pdf>.

credit trading, limit credit trading for alternative fuel vehicles, or prohibit manufacturers of alternative fuel vehicles from receiving and trading credits.

If Congress determines that EVs support the intent of the CAFE program, it could affirm the ability of EV or other alternative fuels vehicle manufacturers to trade and accrue credits. This would allow for more regulatory certainty for manufacturers of these alternative fuel vehicles and their ability to accrue, trade, and sell credits.

Standard Setting

NHTSA has statutory limitations when setting CAFE standards, which Congress might choose to oversee or amend. Statute prohibits NHTSA from considering dedicated fueled vehicles, dual-fueled vehicles operating on alternative fuels, and the trading and transferring of credits when setting standards.¹⁰⁹ NHTSA has taken different approaches to the consideration of EVs in the rulemaking process—such as consideration of EVs in the reference baseline.¹¹⁰ Congress might clarify whether and how alternative fuel vehicles and credit trading may be incorporated in standard setting. Potential approaches include adjusting limitations for these vehicles and credit trading in standard setting, defining the reference baseline in standard setting, or limiting the inclusion of certain alternative fuel vehicles from aspects of the rulemaking process, such as considerations in the reference baseline.

Author Information

Naseeb A. Souweidane
Analyst in Transportation Policy

Disclaimer

This document was prepared by the Congressional Research Service (CRS). CRS serves as nonpartisan shared staff to congressional committees and Members of Congress. It operates solely at the behest of and under the direction of Congress. Information in a CRS Report should not be relied upon for purposes other than public understanding of information that has been provided by CRS to Members of Congress in connection with CRS's institutional role. CRS Reports, as a work of the United States Government, are not subject to copyright protection in the United States. Any CRS Report may be reproduced and distributed in its entirety without permission from CRS. However, as a CRS Report may include copyrighted images or material from a third party, you may need to obtain the permission of the copyright holder if you wish to copy or otherwise use copyrighted material.

¹⁰⁹ 49 U.S.C. §32902(h).

¹¹⁰ NHTSA, "CAFE Standards for Passenger Cars and Light Trucks for Model Years 2027 and Beyond and Fuel Efficiency Standards for Heavy-Duty Pickup Trucks and Vans for Model Years 2030 and Beyond," 89 *Federal Register* 52540, 52786-52787, June 24, 2024; NHTSA, "Resetting the Corporate Average Fuel Economy Program," 90 *Federal Register* 24518, June 11, 2025; and NHTSA, "SAFE Vehicles Rule III for Model Years 2022 to 2031 Passenger Cars and Light Trucks," 91 *Federal Register* 61988.